

Quiz 1

Set: Qualcomm

Ques 1

a. Wafer diameter = 300 mm

$$\therefore \text{Wafer Area} = \pi \left(\frac{300}{2} \right)^2 \text{ mm}^2$$
$$= 70650 \text{ mm}^2$$

$$\text{Yield} = 89\% = \frac{\text{no of working dies}}{\text{dies per wafer}}$$

$$\Rightarrow 0.89 = \frac{415}{\text{Dies per wafer}}$$

$$\Rightarrow \text{Dies per wafer} \approx 466$$

$$\text{Die area} \approx \frac{\text{Wafer Area}}{\text{Dies per wafer}}$$

$$\approx \frac{70650}{466}$$

✓

$$\approx 151.609 \text{ mm}$$

$$\begin{aligned} \text{Cost per die} &= \frac{\text{Cost per die}}{\text{no of working dies}} \\ &= \frac{120}{415} = 0.289 \end{aligned}$$

b. Reference time = 1372s

$$\text{Total clock cycles} = 520$$

$$\text{Instruction Count} = 350$$

$$\text{Avg CPI} = \frac{520}{350} = 1.486$$

$$\text{SPEC Ratio} = \frac{\text{Ref time}}{\text{Ex time}}$$

$$\Rightarrow \text{Ex time} = \frac{1372}{0.8} = 1715s$$

$$\text{Execution time} = \frac{\text{Clock cycles}}{\text{Clock Rate}}$$

$$\Rightarrow \text{Clock rate} = \frac{520}{1715} \\ = 0.303 \text{ Hz}$$

$$\text{Clock period} = \frac{1}{0.303} = 3.3 \text{ s}$$

c. As the execution time is greater than reference time, the cpu is not working correctly

Set : AMD

$$\text{Wafer radius} = 175 \text{ mm}$$

$$\text{Wafer Area} = \pi (175)^2 \text{ mm}^2 \\ = 96208.44 \text{ mm}^2$$

$$\text{Yield} = 92\% = \frac{\text{Working dies}}{\text{Total dies}}$$

Dies per wafer

$$\Rightarrow 0.92 = \frac{\text{No of working dies}}{490}$$

$$\Rightarrow \text{No of working dies} = \frac{0.92 \times 490}{1} = 450$$

$$\text{Die Area} \approx \frac{\text{Wafer Area}}{\text{Dies per wafer}}$$

$$\approx \frac{96208.44}{490}$$

$$\approx 196.34 \text{ mm}^2$$

$$\text{Cost per die} = \frac{\text{Wafer cost}}{\text{Dies per wafer} \times \text{Yield}}$$

$$= \frac{150}{490 \times 92\%}$$

$$\approx 0.33 \$$$

$$b. \text{ Avg CPI} = 4$$

'' 0

Execution time = 1605s

Reference time = 1520s

$$\begin{aligned}\text{SPEC Ratio} &= \frac{\text{Ref time}}{\text{Ex time}} \\ &= \frac{1520s}{1605s} = 0.94\end{aligned}$$

$$\text{Execution time} = \frac{\text{Clock cycles}}{\text{Clock rate}}$$

$$\begin{aligned}\Rightarrow \text{Clock rate} &= \frac{\text{Clock cycles}}{\text{Exec time}} \\ &= \frac{960}{1605} = 0.598 \text{ Hz}\end{aligned}$$

$$\begin{aligned}\text{Clock period} &= \frac{1}{\text{Clock Rate}} \\ &= \frac{1}{0.598} = 1.695s\end{aligned}$$

$$\text{Translation count} = \frac{\text{Clock cycles}}{\text{Clock period}}$$

Instruction count = $\frac{\text{Avg CPI}}{}$

$$= \frac{960}{4}$$

$$= 240$$

c. Execution time greater than
Reference time $\Rightarrow$ not working properly.

Set : Apple

a. Die area = 79.8 mm^2

Dies per wafer = 550

Working dies = 508

Cost per die = $2.75\$$

$$\begin{aligned}
 \text{Wafer Area} &\approx \text{Die Area} \times \text{dies per wafer} \\
 &\approx 79.88 \times 550 \text{ mm}^2 \\
 &= 43934 \text{ mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Yield} &= \frac{\text{Working dies}}{\text{Dies per wafer}} \times 100\% \\
 &= \frac{508}{550} \times 100\% = 92\%
 \end{aligned}$$

$$= \frac{1}{(1 + (\text{Defects/mm}^2 \times \text{Die Area}))^N}$$

$$\Rightarrow \text{Defect/mm}^2 = \frac{\left(\sqrt[N]{\frac{1}{\text{Yield}}} - 1 \right)}{\text{Die Area}}$$

$$= \frac{\sqrt{\frac{1}{0.92}} - 1}{79.8}$$

$$= 0.00053$$

Cost per wafer

$$= \text{cost per die} \times \text{No of working dies}$$

$$= 2.75 \times 508 = 1397 \$$$

b.

$$\text{Execution time} = 1440s$$

$$\text{Overall improvement} = 5$$

$$T_{\text{improved}} = \frac{1440}{5} = 288s$$

$$= \frac{T_{\text{affected}}}{n} + T_{\text{unaffected}}$$

→ Improvement

T_{affected}

$$\begin{aligned}
 \text{factor, } \eta &= \frac{T_{\text{unaffected}}}{T_{\text{improved}} - T_{\text{unaffected}}} \\
 &= \frac{1200}{288 - 240} \\
 &= 25
 \end{aligned}$$

$$\begin{aligned}
 \text{Instruction Count} &= \frac{\text{Clock cycles}}{\text{Avg CPI}} \\
 &= \frac{\text{Execution time} / \text{Clock period}}{\text{Avg CPI}} \\
 &= \frac{1440 / 2}{4} = 180
 \end{aligned}$$

Performance

$$\text{after improvement} = \frac{1}{T_{\text{improved}}}$$

$$= \frac{1}{288}$$

$$= 0.0034$$

c. make the common case first
so that it would have the
most effect.

Set: Media Tek

$$\text{Die area} = 140.79 \text{ mm}^2$$

$$\text{Cost per die} = 1.4 \$$$

$$\text{Working die} = 375$$

$$\text{Dies per wafer} = 390$$

$$\text{Wafer Area} \approx \text{Die Area} \times \text{Dies per Wafer}$$

$$\Rightarrow \pi r^2 \approx 390 \times 140.70 \text{ mm}^2$$

$$\Rightarrow \pi r^2 \approx 54908.1 \text{ mm}^2$$

$$\Rightarrow r = \sqrt{\left(\frac{\quad}{\pi} \right)} = 122.203 \text{ mm}$$

$$\therefore \text{Wafer diameter} = 2 \times r$$

$$= 244.406 \text{ mm}$$

$$\text{Yield} = \frac{\text{Working dies}}{\text{Dies per wafer}} \times 100\%$$

$$= \frac{375}{390} \times 100\%$$

$$= 94\%$$

$$\text{Defects / mm}^2 = \frac{\left(\sqrt{\frac{1}{\text{Yield}}} - 1 \right)}{\text{Die Area}}$$

$$= \frac{\sqrt{\frac{1}{140.79} - 1}}{140.79}$$

$$= 0.00022$$

$$\text{Cost per wafer} = \text{Cost per die} \times \text{Working die}$$

$$= 1.4 \times 375 = 525$$

b. Execution time = 950s

Overall improvement = 3

$$T_{\text{improved}} = \frac{950}{3} = 316.667$$

$$= \frac{T_{\text{affected}}}{n} + T_{\text{unaffected}}$$

$$i) \text{ Improvement factor, } n = \frac{T_{\text{affected}}}{T_{\text{improved}} - T_{\text{unaffected}}}$$

$$= \frac{720}{316.67 - 230}$$

$$= 8.307$$

$$\begin{aligned}
 \text{Avg CPI} &= \frac{\text{Clock cycles}}{\text{Instruction Count}} \\
 &= \frac{\text{Execution time} / \text{Clock period}}{\text{Instruction Count}} \\
 &= \frac{0.50 / 1.75}{320} = 1.696
 \end{aligned}$$

Performance

$$\text{before improvement} = \frac{1}{\text{Exec time}}$$

$$= \frac{1}{0.50}$$

$$= 0.0010$$

C. Amdahl's law.

Performance after improvement

$$= \frac{1}{T_{\text{improved}}} = \frac{1}{31.6.67} \\ = 0.003$$

Performance increased

$$= (\text{After} - \text{Before}) \\ = 0.002$$